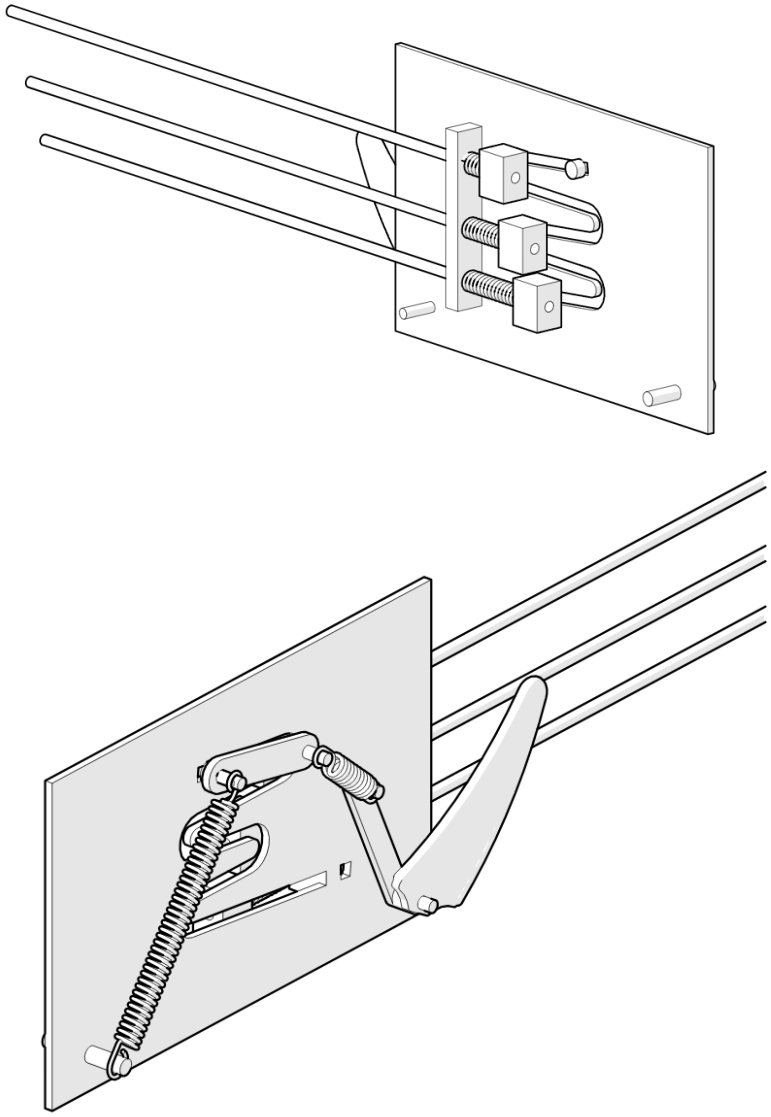
Design and Dynamic Analysis of a Spring-Loaded Trigger Mechanism

Mechanism Engineering Report

March 14, 2026



Abstract

This document details the design, kinematics, and dynamic simulation of a mechanical trigger assembly. The analysis includes a breakdown of the sub-components, an overview of the working principles, and an evaluation of the torque required to actuate the mechanism under a 0.5 kg load based on Simscape Multibody simulations.

1 Working Mechanism Illustration

The mechanism relies on a linkage system controlled by two primary tension springs to govern the actuation and reset phases. When the trigger is pulled, it rotates around its pivot, pulling the connecting link and the sliding pin forward against the resistance of the return spring.

The physical layout of the components is detailed below from both the front and rear perspectives for clear part labeling.

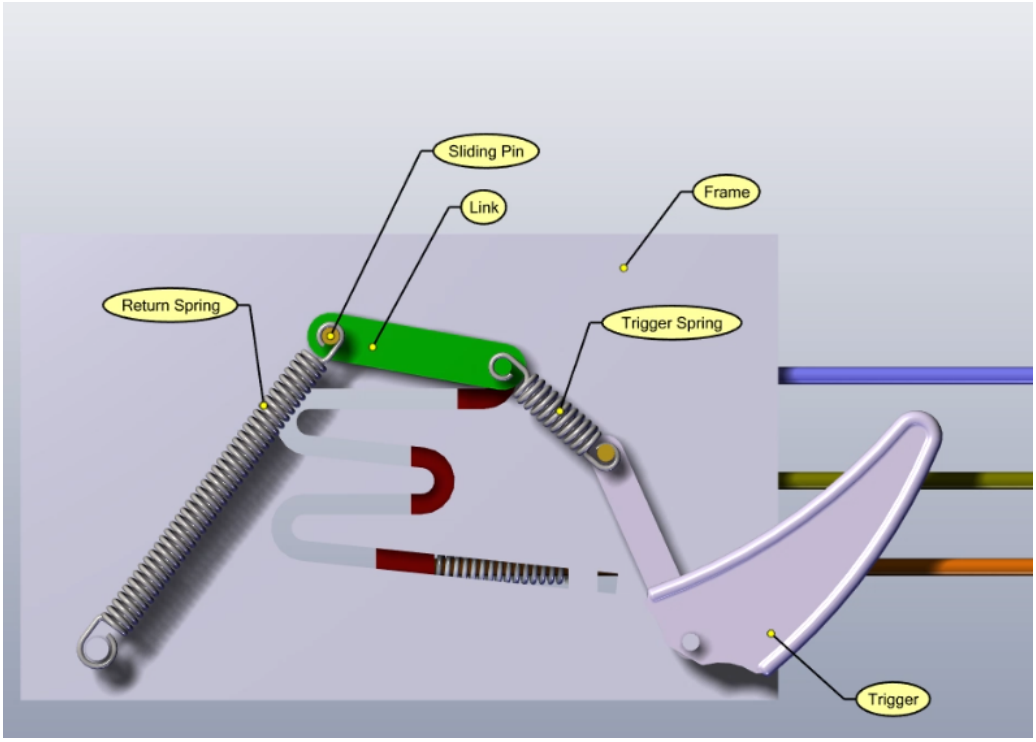


Figure 1: Front view of the mechanism.

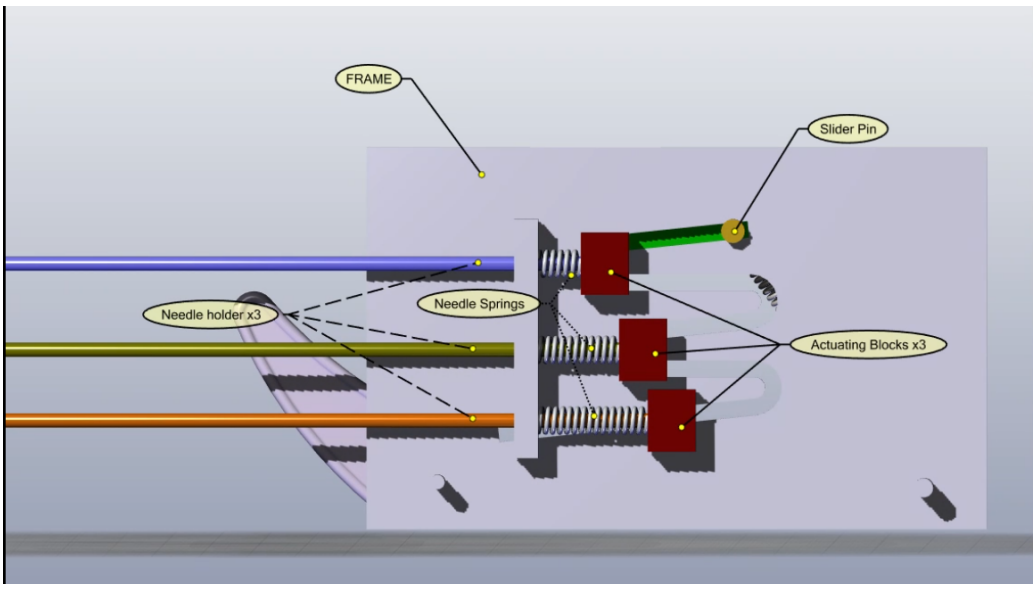


Figure 2: Rear view of the mechanism.

The trigger spring maintains tension between the frame and the trigger to provide a crisp reset, while the heavier return spring is responsible for driving the sliding pin back to its resting state once the sear/trigger is released.

2 Parts List and Bill of Materials

The mechanism consists of a simplified kinematic chain with rigid bodies and force elements. The primary components are detailed in Table 1.

Table 1: Primary Components of the Trigger Assembly

Item No.	Part Description	Quantity
1	Frame	1
2	Trigger	1
3	Link	1
4	Sliding Pin (Slider Pin)	1
5	Trigger Spring	1
6	Return Spring	1
7	Needle Holder	3
8	Needle Spring	3
9	Actuating Block	3
10	Pivot Pins	3

3 Operational Phases

The actuation cycle of the mechanism is broken down into six distinct phases, illustrating the progression of the linkages throughout a complete pull and reset cycle.

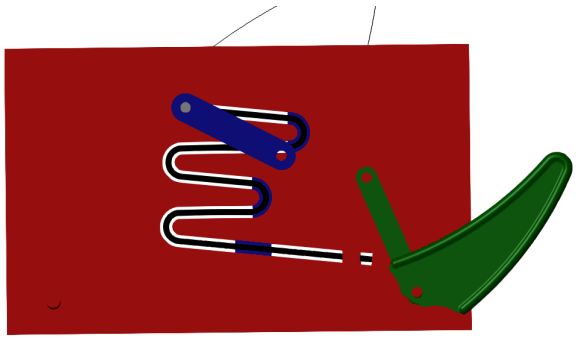


Figure 3: Phase 1: Initial Resting state.

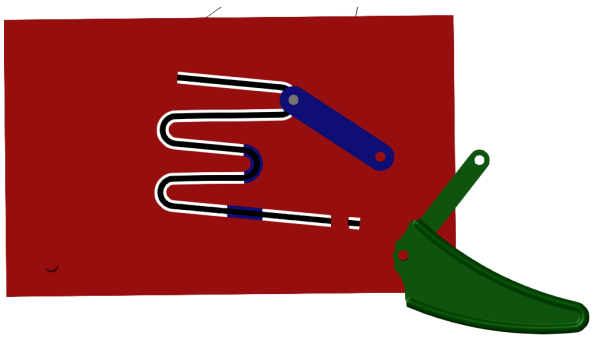


Figure 4: Phase 2: pull and 1st needle actuated.

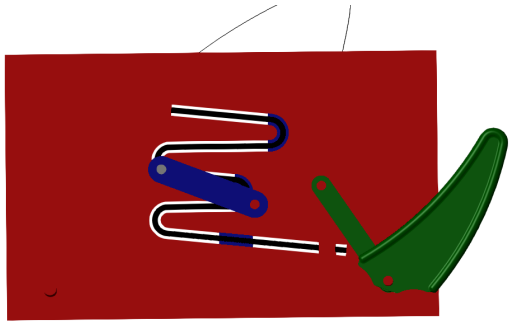


Figure 5: Phase 3: Release and 2rd resting state.

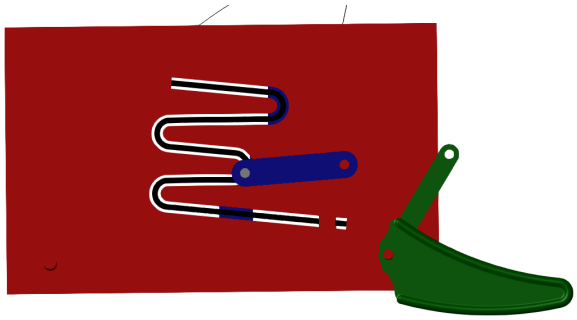


Figure 6: Phase 4: Pull and 2nd needle actuated.

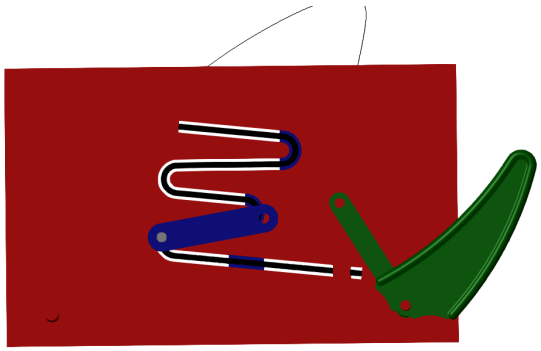


Figure 7: Phase 5: Release and 3rd resting state.

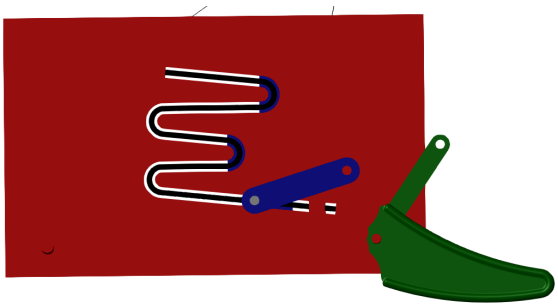


Figure 8: Phase 6: Pulls 3rd needle actuated

4 Dynamic Analysis and Torque Requirements

A rigid-body dynamics simulation was conducted in MATLAB/Simscape to evaluate the holding and actuation torque at the trigger pivot. The design target for the mechanism was a comfortable finger-pull load of approximately 0.5 kg.

Given a mass $m = 0.5$ kg, the applied force F is calculated as:

$$F = m \cdot g = 0.5 \cdot 9.81 \approx 4.9 \text{ N} \tag{1}$$

Assuming a standard trigger moment arm $r = 0.03$ m, the theoretical required torque τ is approximately 0.15 Nm (150 Nmm).

The dynamic simulation results capture both the kinematic motion and kinetic requirements over multiple actuation cycles, which are broken down in the following sections.

5 Trigger Position vs. Time

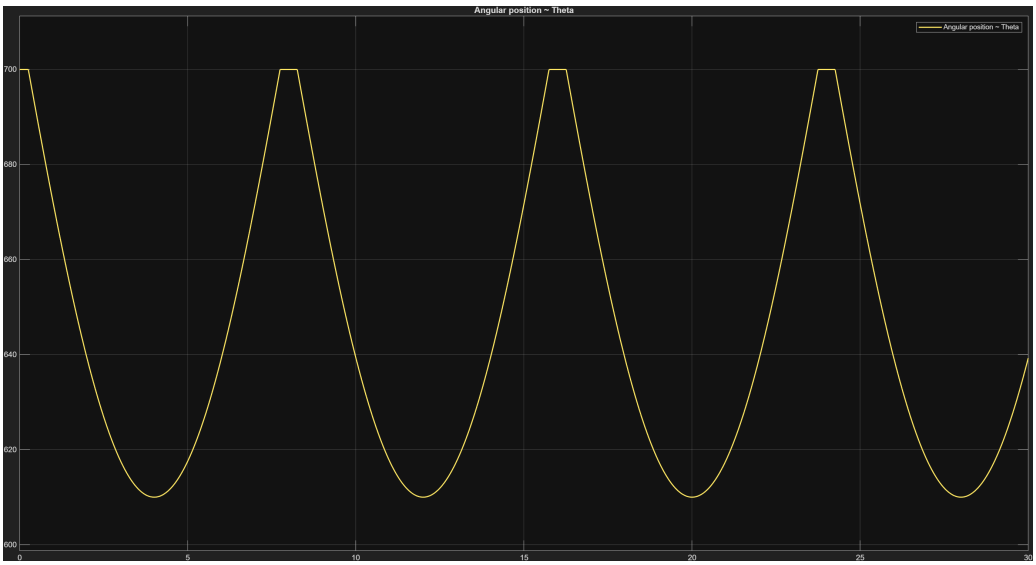


Figure 9: Simscape simulation output showing the kinematic position of the trigger over time.

6 Trigger Torque vs. Time

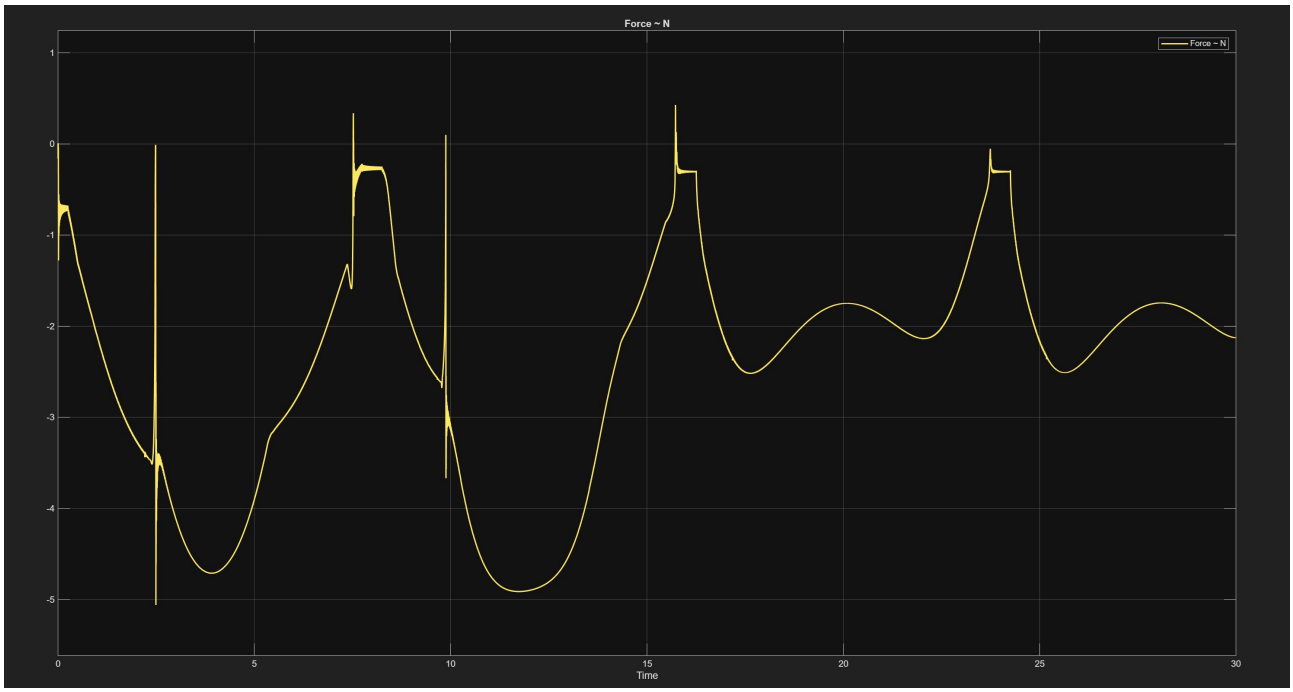


Figure 10: Simscape simulation output showing the required actuation torque over a multi-cycle test.

The torque graph exhibits transient spikes during the initial pull and reset phases, indicating sudden impacts or hard-stop collisions within the joint limits. Depending on the simulation's base unit scale (Nm vs Nmm), these spikes (peaking at magnitude 9) suggest either excessive inertia in the CAD model properties or a lack of sufficient damping coefficients in the Revolute Joint configuration.

7 Conclusion and Recommendations

This spring-loaded trigger mechanism provides a simple solution that is highly advantageous due to its ease of manufacturing and low overall production cost. While the kinematic linkages function as intended to drive the sliding pin, the system requires optimizing the springs' stiffness to provide a better, more comfortable trigger feel. Furthermore, the performance of the sliding roller may be affected by the physical path it moves in, requiring careful consideration of tolerances and friction. Moving forward, it is also recommended to mitigate the severe transient torque spikes observed in Figure 10 by adjusting damping coefficients and verifying the torque output scaling in the dynamic simulation.